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Methodological Reflections on PISA'S Creative Thinking Assessment

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ABSTRACT

The Programme for International Student Assessment (PISA) introduced creative thinking as an innovative domain in 2022. This paper examines the unique methodological issues in international assessments and the implications of measuring creative thinking within PISA's framework, including stratified sampling, rotated form designs, and a distinct scoring metric. Cross-cultural considerations and data quality issues underscore the need for critical evaluation of cultural equivalence and data quality, especially at the individual educational system level. The paper also highlights the limitations of causal inference and provides guidance and resources for handling sampling weights, plausible values, and multilevel structures. By reflecting on PISA's history and design, this work offers insights for researchers navigating the complexities of innovative domain assessment to ensure valid interpretations.

1 | Introduction

The Programme for International Student Assessment (PISA), initiated by the Organization for Economic Co-operation and Development (OECD) in 2000, evaluates educational system quality by measuring 15-year-old students' performance in reading, mathematics, science, and an innovative domain that changes in each cycle. Originally conducted on a 3-year cycle, excepting a one-year delay from 2021 to 2022 due to the COVID-19 pandemic, the study will transition to a 4-year cycle beginning in 2029. The creative thinking innovative domain, measured in 2022, is the focus of this article. PISA focuses on assessing students' ability to apply their knowledge to real-world problems, thus providing valuable insights into the effectiveness of education in participating countries and economies. The results offer a snapshot of students' skills and the context of their knowledge.

While other contributions in this special issue delve into the conceptual and educational dimensions of creative thinking, this article addresses the methodological challenges and opportunities unique to this domain. Specifically, it focuses on how PISA's design and analytic choices—sampling, assessment frameworks, and achievement estimation—shape the validity and utility of creative thinking data. These insights are particularly relevant for researchers seeking to analyze the data effectively or draw inferences across diverse educational contexts. By situating the discussion within PISA's broader framework, this article also reflects on the innovative domain's role in expanding the scope of large-scale assessments. Creative thinking represents a critical skill for addressing complex global challenges, and understanding its measurement in PISA offers valuable lessons for both future research and policy design.

2 | General Design Features of PISA

Because of the aims of international large-scale assessments (ILSAs) like PISA—to measure what a representative sample of students knows and can do in broad content areas—the designs of these studies feature several idiosyncrasies that are important for researchers interested in using the resultant data to consider. We briefly describe each of these topics and provide references for readers who are interested in a deeper understanding of each topic.

As noted above, PISA measures 15-year-olds in specified content domains. A secondary but important part of the PISA data collection is a set of questionnaires that are administered to students, school leaders, and, depending on the participating educational system's preferences, parents and teachers. These questionnaire data serve as an important basis from which to draw inferences around predictors of achievement from many aspects of a student's life. As we describe subsequently, student responses to the questionnaires also serve as an important component of the model used to estimate population-level achievement.

2.1 | Sampling Design

PISA employs a two-stage stratified sampling design to ensure that the results are representative of the 15-year-old student population in each participating country (OECD 2024a). Stratified sampling involves dividing the population into subgroups (or strata) to ensure that each subgroup is adequately represented in the sample, which helps to improve the precision of the results (Rust 2014). An example of one such stratification variable could be school type, with the types being public and private. Geographic region is another stratification example. Then, within each stratum, the first stage of sampling involves selecting schools based on a probability proportional to their size (Lohr 2019; Thompson 2012), where larger schools are selected with higher probability. In the second stage, 15-year-old students within these schools are randomly selected to participate. For the PISA 2022 cycle, the typical sample size comprised approximately 150 schools and 6000 students per participating country, although these numbers can vary depending on the structure of the country's education system and the aims of a given country (ensuring a representative sample of indigenous students, for example). Within each school, the sample size will differ based on the assessment platform. For instance, countries using computer-based assessments select 42 students per school, while those using paper-based assessments select 35 students per school.

2.2 | Assessment Design

PISA's assessment design ensures that not all students take all items. Instead, a sample of students answers a sample of items according to a sophisticated design (Gonzalez and Rutkowski 2010; Rutkowski et al. 2010). This method gathers comprehensive achievement data without overburdening individual students. Due to its complexity, we do not provide a comprehensive explanation of PISA's assessment design. Instead, we offer a simple overview for newcomers to these concepts. Although some countries, particularly low-income countries, continue to take PISA as a paper-based assessment, most countries administer PISA via a computer platform. Additionally, a multi-stage adaptive test (MST; Yamamoto et al. 2019; Yan et al. 2014) design has been implemented in recent cycles. An MST design enables limited adaptation of a test's difficulty based on an examinee's proficiency. Unlike a fully computerized adaptive test (CAT), where adaptation occurs at the item level, MST adapts at the module or testlet level. A module is defined as a group of items that always appear together in a block or cluster.

An example of an MST is illustrated in Figure 1, which depicts a three-stage MST structure, collectively referred to as a panel. This panel outlines the various pathways an examinee may follow. Examinees start in stage 1 with a core or routing testlet. Based on their performance in this core testlet, they are directed, via a decision rule, to either an easier or more difficult stage 2 testlet. Further routing into stage 3 testlets depends on performance in the preceding stages. In total, each examinee receives three blocks of non-overlapping items. Although this description pertains to a single panel, an MST can—and in PISA, does—consist of multiple panels. MST was implemented in PISA 2018 for the reading domain, in 2022 for the math domain, and is planned for the science domain in 2025.

For domains that are not adaptive, a *rotated* form design is used. This sort of design is referred to as item-sampling (Lord 1962) or, more commonly in modern literature, as multiple-matrix sampling (Shoemaker 1973). In multiple-matrix sampling, items are distributed into non-overlapping clusters, which are then distributed across forms according to a design (Gonzalez and Rutkowski 2010). Cluster distribution is done in such a way as to ensure sufficient overlap to allow for linking across forms and sufficiently precise estimates of population and sub-population achievement (Mislevy et al. 1992; von Davier and Sinharay 2014). In PISA 2022, 28% of the total student sample participated in the creative thinking assessment (OECD 2023). Among these students, 86% (24 out of the 28 percentage points) also took a math assessment, while the remaining 14% (4 out of the 28 percentage points) took creative thinking along with a reading or science assessment. We illustrate the basic math/creative thinking design in Table 1. Readers are invited to consult the PISA 2022 framework or technical report for full details (OECD 2023, 2024d). Table 1 shows that there are four clusters for each form and that either the first two or last two clusters feature creative thinking

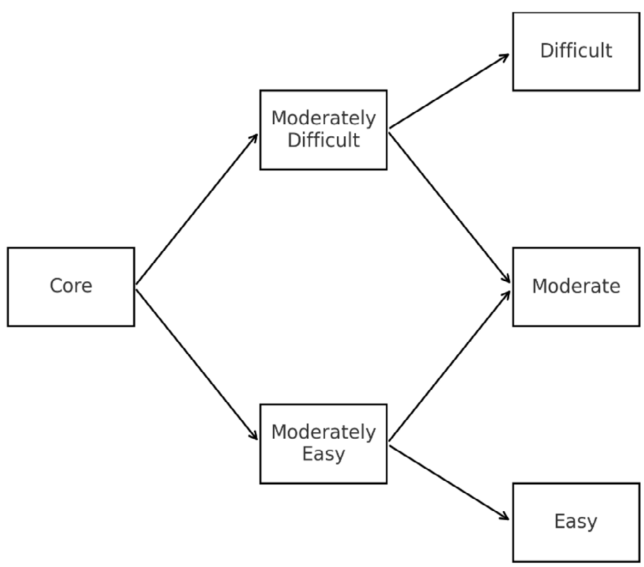

FIGURE 1 | Example of a 3-stage MST design.

TABLE 1 | PISA 2022 design for math and creative thinking.

Form	Cluster 1	Cluster 2	Cluster 3	Cluster 4
37	M(adaptive)		CT1	CT2
38	M(adaptive)		CT2	CT3
39	M(adaptive)		CT3	CT4
40	M(adaptive)		CT4	CT5
41	M(adaptive)		CT5	CT1
42	CT2	CT4	M(adaptive)	
43	CT3	CT5	M(adaptive)	
44	CT4	CT1	M(adaptive)	
45	CT5	CT2	M(adaptive)	
46	CT1	CT3	M(adaptive)	

Abbreviations: CT*x*, creative thinking where x = item cluster 1 to 5; M(adaptive), math domain delivered via an MST design.

items, which are arranged into non-overlapping blocks, from 1 to 5. Here, we can see that the rotation design is such that creative thinking item block 1 appears in all four cluster positions and with three other creative thinking blocks: with cluster 4 in Form 37, with cluster 5 in Form 41, with cluster 4 in Form 44, and with cluster 3 in Form 46. The design is balanced in that cluster 1 (and all creative thinking clusters) appears in all four positions. And each item cluster is administered before and after mathematics. Notable is that each student sees just 2 of the 5 total creative thinking clusters. And the creative thinking domain is non-adaptive.

2.3 | Estimating Achievement Distributions

The assessment design used in PISA requires that achievement in measured domains is estimated at the population and subpopulation levels using highly specialized methods that are related to multiple imputation for missing data (Khorramdel et al. 2020; Mislevy et al. 1992; Rubin 1987). We do not cover these methods in detail; however, interested readers can consult several introductory resources on this topic (Gonzalez and Rutkowski 2010; Rutkowski et al. 2010; von Davier et al. 2006, 2009). Essentially, achievement is assumed missing for every examinee, and these missing values are imputed using a complex, highly specified imputation or conditioning model that includes responses to all student background information and other important demographic information. This approach is analogous to multiple imputation in the missing data literature (Rubin 1987). Further, correlations among the tested domains are estimated as part of a complex latent variable model. For example, latent correlations among math, science, and reading are estimated. As we describe subsequently, creative thinking was handled in a slightly different way. We describe the implications of this treatment later.

2.4 | Cross-Cultural Considerations

Beyond the unique sampling and assessment design aspects, the cross-cultural nature of PISA is an important methodological feature. In the 2022 cycle, PISA featured 81 educational systems, representing a highly diverse range of countries and economies with varied linguistic, cultural, economic, and geographic contexts. These variations can influence how students interpret the assessed domains and respond to test questions, beyond differences in their content knowledge (Millsap 2011; van de Vijver and Leung 1997). For example, within a single country, students from urban areas might have more exposure to diverse educational resources and extracurricular activities compared to their peers in rural areas. Further, their environmental context will likely influence what they know about their world. This can influence student interpretations and responses to test questions that involve real-world scenarios. An urban student might find a question about public bus schedules straightforward due to personal experience, while a rural student might struggle to relate to the context, even if both students possess similar content knowledge. Such contextual factors are important to consider when analyzing and interpreting PISA results.

3 | What These Issues Mean for Empirical Researchers

3.1 | Sampling Design Implications

Because students are sampled according to a complex design and inferences are intended at the population level, it is important to incorporate sampling weights and appropriate methods of variance estimation (Efron and Gong 1983; Thompson 2012). Sampling weights are crucial in complex survey designs like PISA as they ensure that the sample accurately represents the population (Pfeffermann 1993; Rust 2014). These weights adjust for the different probabilities of selection and help mitigate the effects of non-response and other biases. Essentially, a sampling weight is a number assigned to each case in the data set, which represents the inverse of the probability of selection for the case. This is analogous to $\frac{1}{n}$ in the standard mean formula (e.g., $\frac{1}{n}\sum x_i$) where each observation, x_i is implicitly selected with a probability of $\frac{1}{n}$ and, therefore, in a weighting scheme, would receive a weight of n, indicating how much that case should count in statistical analyses to produce unbiased estimates in the population.

Additionally, resampling methods, such as the jackknife or bootstrap (Efron 1982), are often used to estimate the variance of population parameters in complex survey designs. The usual variance formula is not appropriate in this context because it does not account for the complex survey design, which can lead to underestimation of the variance. These resampling methods help in quantifying the uncertainty of the estimates by accounting for the complex design, which is critical for making valid statistical inferences. Specifically, PISA employs the jackknife method with balanced repeated replicates (BRR) to estimate variance. Traditional variance formulas are not sufficient in this context, as they do not account for the complex survey design, which can lead to underestimation of the variance. The jackknife with BRR addresses this by providing more accurate variance estimates.

We also note that the stratified sampling design ensures that demographic groups of interest are adequately represented.

For researchers who are interested in analyses that include stratification variables, either as a variable of interest or as a control variable, be sure to consult the technical report. Annex Table A.6.3 on pages. 127–130 provides a complete account of stratification variables used in PISA 2022 (OECD 2024d).

Beyond incorporating sampling weights and appropriate variance estimation, it can also be useful to consider the multilevel structure of the data. PISA data is inherently multilevel, with students nested within schools and, depending on the nature of the analysis at hand, within regions or countries. Ignoring this structure can lead to underestimated standard errors and inflated Type I error rates[1] (Snijders and Bosker 2012). Multilevel modeling allows researchers to partition the variance at different levels and appropriately account for the dependencies in the data. This approach often requires the use of different types of sampling weights at each level—weights that reflect the probability of selection of schools and students within those schools (Asparouhov and Muthén 2006; Pfeffermann et al. 1998). By correctly modeling the hierarchical nature of the data and using appropriate sampling weights, researchers can obtain more accurate and generalizable findings (Kim et al. 2014).

3.2 | Implications of Achievement Estimation Methods

PISA estimates achievement by producing multiple plausible values (PVs) for each student. Historically, older cycles of PISA have produced 5 PVs per student (e.g., OECD 2014). A shift to generating 10 PVs per student began with the 2015 cycle (OECD 2017). These PVs are drawn from the posterior distribution of the latent trait based on a student's responses to the test and, via an imputation model, to the background questionnaire and other select demographic data (Khorramdel et al. 2020; Rutkowski et al. 2010; von Davier et al. 2009). Analysts must use all PVs in their analyses to obtain unbiased estimates of population parameters and, along with proper variance estimation methods, accurate standard errors. In particular, a given analysis or model should be fitted once to each of the PVs, and the resulting parameter estimates and standard errors should be combined according to Rubin's rules (Rubin 1987, p. 198). Improperly handling PVs (using just one or averaging them) can lead to biased estimates and underestimation of the uncertainty associated with the results (Rutkowski et al. 2010). Historically, the PISA scale is standardized to have a mean of 500 and a standard deviation of 100 in the initial PISA cycle. This standardization facilitates comparisons over time and across different populations, ensuring that changes in performance can be attributed to actual differences rather than variations in the measurement scale.

3.3 | Implications of Cross-Cultural Measurement Differences

Prior to 2015, models used to estimate population achievement assumed identical parameters for math, science, and reading items across all countries. Items showing strong deviations from this assumption—using ad hoc methods—were labeled "dodgy" and discarded or not used for specific countries. In 2015 and onward, procedures changed to quantify *differential item functioning* (DIF) and allow for item-by-country interactions (Glas and Jehangir 2014; Oliveri and von Davier 2011, 2014). DIF analysis helps identify items that function differently across diverse groups of students, despite having the same underlying ability. By flagging and investigating these items, the OECD ensures that the test is as fair as possible for all participating countries. This involves statistical analyses and expert reviews to determine whether cultural biases are present in the test items or background questions. In a similar fashion, constructs developed from the background questionnaires (i.e., *motivation*) were also formally investigated for DIF (Buchholz and Hartig 2017). Importantly, questionnaires are more vulnerable to response-style bias, such as socially desirable responding (choosing an answer that projects a favorable image of the respondent), acquiescent response style (a tendency to agree), and extreme response style (a tendency to use the endpoints of the scale, such as *strongly agree* or *strongly disagree*) (Johnson et al. 2011).

Although much work goes into ensuring a reasonable degree of equivalence for achievement domains and other non-achievement constructs, the OECD necessarily focuses on an overall, international level. As such, researchers who use PISA data should evaluate, where possible, the equivalence assumption using multiple-group invariance methods such as multigroup confirmatory factor analysis or analogous item response theory methods (Buchholz and Hartig 2017; Millsap 2011; Millsap and Yun-Tein 2004). We note, however, that the complexity of detecting and correcting for DIF in achievement domains makes this sort of investigation most practical for non-achievement constructs that are derived from the background questionnaires.

3.4 | Software

To correctly analyze PISA and other ILSA data, we recommend that users familiarize themselves with several free or paid resources. First, in *R*, researchers can use several packages for conducting basic, single-level analyses that correctly incorporate sampling weights, handle plausible values, and estimate variance appropriately. Although not exhaustive, *RALSA* (Mirazchiyski 2024) and *EdSurvey* (Bailey et al. 2019) are user-friendly options with good documentation. For multilevel analyses, we recommend *WeMix* (Bailey et al. 2023) in conjunction with Huang's (2024) supplementary functions for handling plausible values, both of which are available in the *R* environment. We also recommend *M*plus (Muthén and Muthén 1998), which is a flexible platform for conducting simple, single-level analyses and for fitting multilevel and latent variable models. For investigating measurement invariance, users can consult a detailed tutorial for *R* and *M*plus (Svetina et al. 2020) using a confirmatory factor analytic approach. For an item response theory approach, researchers can use *mirt* (Chalmers 2012) or *TAM* (Robitzsch et al. 2024), both of which are in *R* and can estimate the root-mean-squared deviation (OECD 2024b; Oliveri and von Davier 2011, 2014) used by the OECD for detecting DIF.

4 | Creative Thinking in Particular

In what follows, we describe several design, implementation, analysis, and data quality factors that are specific to the creative thinking domain in PISA. We offer some perspective on analyzing these data for your own research purposes and what to be aware of as you use these data. We note here that 64 educational systems participated in the creative thinking assessment, and the mode of administration was exclusively computer-based (OECD 2024d, p. 234).

4.1 | Assessment Design for Creative Thinking

As previously noted, the proportion of students who took the creative thinking portion of PISA was 28%. This contrasts with the major domain in 2022—mathematics—where 94% of sampled students took the math portion of the test, and 39% of the sample took the reading and science domains, respectively. The number of items in the creative thinking domain is also substantially different from the core domains. For example, there were 197, 234, and 115 reading, math, and science items, respectively. By comparison, the creative thinking domain featured 36 total items divided into five non-overlapping clusters (OECD 2024d, p. 77), which were assembled into forms as shown in Table 1.

4.2 | Scaling and Scoring for Creative Thinking

The results of the scaling analysis for creative thinking resulted in 4 of the 36 items being dropped, for a total of 32 items. See Chapter 14 of the PISA Technical Report for details. Further, the usual scoring metric that follows a normal distribution with a mean of 500 and a standard deviation of 100 was determined to be inappropriate for the creative thinking scale. This was partly due to the small item pool and that not much information was available for the low end of the creative thinking achievement continuum. Instead, the creative thinking plausible values are converted into a score that reflects the "expected number correct on a hypothetical form made up of all the items in the creative thinking item pool, given the proficiency level that the plausible value represents" (OECD 2024d, p. 299). This led to a scale with a minimum of 0 and a maximum of 60. The OECD average on this scale was 33 points with a standard deviation of 11 (OECD 2024c).

In addition to a different reporting metric, the scoring model used to produce achievement scale scores differed for creative thinking. As noted previously, the latent correlations for math, science, and reading are freely estimated as part of the scoring model. In contrast, the correlation between creative thinking and the core domains was estimated separately in a subsequent analysis. In this analysis, the previously estimated latent correlations among the major domains were fixed at their earlier values. Plausible values for creative thinking were then drawn for each student. This adjustment was done to preserve the historic PISA scale for the major domains and to avoid any impact on their estimated values that could be attributed to the much shorter and lower reliability creative thinking scale. In comparison to the core domains, where the latent correlations among math, reading, and science are averaged across countries, between 0.79 and 0.85, the correlation between creative thinking and the core domains is between 0.67 and 0.68 (OECD 2024d, pp. 309–310). The reliability of creative thinking is also somewhat lower than the core domains. International average reliability is 0.80 for creative thinking and ranges between 0.86 and 0.90 for the core domains, the values of which are closer to math subscales, like *quantity* or *space and shape* (pp. 306).

4.3 | What Does This Mean in Practice?

Researchers who plan to use the creative thinking domain in their own research should inspect the scale reliability values for countries of interest. The values can be found in "Annex 14. A" part of the technical report. Researchers should also take care to correctly report the differently scored achievement values for creative thinking and recognize these scores as qualitatively different than the standard 500/100 scale used in other PISA domains. We note, however, that there are still multiple values for each examinee and that they should be handled in the same way as plausible values (each analysis should be repeated once for each plausible value and appropriately combined as previously described). Finally, we draw readers to figure 14.9 on page 298 of the PISA Technical Report, where a plot of item difficulty relative to the empirical and theoretical achievement distribution is also presented. Notable here is that there are no items below the scale score of 10, again highlighting the fact that there is little information about the lowest performers on this scale. We also refer readers to the shape of the empirical distribution, which adheres less to a normality assumption than the other domains (see, for example, figures 14.6 to 14.8). An implication of this relative deviation from normality is that results from any analysis that assumes normally distributed variables can and should be formally evaluated with standard methods, and associated results should be interpreted with care.

5 | Reflections on Innovative Domains

Creating robust, well-developed domain frameworks for ILSAs is complex and iterative, often requiring multiple cycles to refine. This challenge is not unique to PISA but has been a consistent feature of ILSAs since their inception in the late 1950s (Brown 1996; Husén and Postlethwaite 1996). The difficulty lies not only in establishing a shared understanding of what to measure internationally but also in designing an instrument capable of capturing these constructs across diverse educational systems and cultural contexts. These obstacles highlight the dual demands of technical rigor and conceptual clarity. The history of ILSAs provides useful context for these challenges. For example, the First International Mathematics Study (FIMS), one of the earliest ILSAs, encountered difficulties in defining its framework due to the varied curricular priorities and pedagogical approaches across participating countries (Brown 1996; Husén 1979). Reaching consensus required iterative dialogue and numerous compromises on defining the domain and how it was measured. Even in well-established domains like mathematics, defining an assessment framework has never been straightforward or without controversy (Ross and Genevois 2006).

PISA's phased approach to framework development further illustrates this complexity. Since its launch in 2000, PISA has assessed reading, mathematics, and science, but only the reading domain was fully developed in the initial cycle. Mathematics and science were introduced as minor domains and only reached full development in subsequent cycles—mathematics in 2003 and science in 2006. The OECD now revises a domain's framework each time it serves as the major domain, ensuring updates to construct definitions, operationalization, and implementation to maintain contemporary relevance. This process requires significant time and effort to refine constructs, align them with diverse educational goals, pilot items, and ensure psychometric robustness. While some have questioned the quality of PISA and other ILSA results (Goldstein 2013), they have largely been accepted by academic and policy communities as valid indicators of student achievement in assessed systems (Hopfenbeck et al. 2018).

The challenges of developing assessment frameworks for new, innovative domains, such as creative thinking, are even more pronounced. Unlike mathematics, reading, and science, which benefit from multiple cycles of refinement, innovative domains are typically assessed only once, making it difficult to achieve the same level of conceptual and technical rigor. For example, the development of a framework for global competence in PISA 2018 faced significant challenges. As discussed in Engel et al. (2019), efforts to operationalize global competence encountered substantial hurdles, including a lack of a universally accepted definition. The resulting framework reflected a narrow interpretation of global competence, omitting critical dimensions such as values and actions. Instead, it assessed what the OECD termed "global understanding" – a proxy that fell short of capturing the full scope of the construct. While the OECD's effort was ambitious, it also highlighted the inherent difficulties in measuring such broad, multidimensional constructs internationally.

These challenges raise a critical question for analysts and policymakers: in the context of innovative domains such as creative thinking, should perfection be the enemy of the good? That is, even if an assessment does not meet the same psychometric and conceptual standards as PISA's legacy domains, can it still provide useful insights? The OECD, in its efforts to validate international measures, necessarily incorporates country-specific considerations, recognizing that educational systems differ markedly in curricular emphases, cultural contexts, and policy priorities. However, no international assessment can fully capture the intricate nuances of each participating country. Instead, these domains must strike a balance between universal applicability and local relevance—a compromise that inevitably leaves some gaps.

This reality necessitates careful reflection by secondary analysts of the creative thinking scale. They must critically evaluate the compromises inherent in the creative thinking assessment design and determine whether the resulting construct is sufficiently robust for valid and meaningful interpretation within their unique contexts. The question is not just whether the scale functions well at the global level but whether it meets the minimum thresholds necessary for valid secondary analysis. As an innovative domain in PISA, creative thinking features a one-off development and administration, making it unlikely to reach the same level of conceptual and technical rigor as legacy domains like mathematics, science, and reading. Nevertheless, creative thinking may still be valuable, even if somewhat limited in its insights. As such, it is essential to engage with the frameworks and technical report and to carefully qualify and explicitly communicate any relevant limitations that could potentially undermine the validity of inferences.

5.1 | A Non-Exhaustive List of Points to Consider

To ensure valid interpretation of any analysis that uses the creative thinking scale, we advocate for an argument-based approach (Kane 1992). Although not exhaustive, here are some essential steps researchers should take:

- **Understand the OECD's definition**: Researchers should carefully review the PISA creative thinking framework (OECD 2023) to ensure that their conceptualization of creative thinking aligns with the OECD's operational definition. Just as the global competence framework in PISA 2018 faced definitional challenges, the creative thinking domain may not fully capture all aspects of creative thinking as defined by various theoretical traditions. Researchers should assess whether PISA's framework meaningfully aligns with their intended use of the data.

- **Assess construct alignment**: Given PISA's broad international applicability, researchers must critically evaluate how the construct definition and operationalization align with their specific needs. While PISA's framework is designed to be generalizable across diverse education systems, this generalization can introduce limitations. For instance, different cultures may conceptualize or approach creative thinking differently, leading to variations in how students engage with the assessment. Recognizing such potential discrepancies is crucial when interpreting results.

- **Evaluate data suitability**: Researchers should thoroughly review the technical report (OECD 2024d), which details how the data were collected, scaled, and analyzed. This report provides essential information on data quality, including response rates, item functioning, and reliability estimates.[2] Given that data quality varies substantially across educational systems, researchers should be cautious about generalizing findings from systems where data quality differs significantly from the international norm.

- **Recognize causal inference limitations**: The low-stakes, observational, and cross-sectional nature of PISA poses significant challenges for causal inference. As highlighted in Rutkowski et al. (2024) and Rutkowski (2016), even sophisticated quasi-experimental methods face obstacles due to the nature of the data. Educational systems differ in policies, curricula, and student populations, which can violate key assumptions such as homogeneity, ignorability, and the absence of hidden treatment variation.

Consequently, researchers should exercise caution when making causal claims, particularly about the effects of educational interventions or policy decisions.

- **Account for measurement limitations**: The development of an internationally appropriate framework for creative thinking posed several methodological challenges. Unlike established domains such as math, reading, and science, which have undergone multiple iterations of refinement, creative thinking was assessed for the first time in PISA 2022. As a result, its psychometric properties may not be as robust as those of legacy domains. Researchers must carefully examine item-level functioning, reliability estimates, and measurement invariance across groups before drawing strong conclusions.

In this paper, we offer general guidelines for conducting analyses that account for the complex research design of the PISA assessment. In particular, we advise researchers to account for sampling weights, potentially the nested structure, and plausible values to ensure that their findings are, within reason, robust and generalizable. Ignoring these design complexities can lead to biased estimates and incorrect inferences, undermining the validity of the conclusions. However, careful and thoughtful analysis has the potential to lead to important insights at both the academic and policy levels.

6 | Conclusion

As international large-scale assessments continue to evolve, innovative domains like creative thinking offer valuable opportunities to expand our understanding of student competencies beyond traditional academic subjects. However, these domains also come with distinct methodological challenges that require careful consideration. While the creative thinking scale in PISA 2022 does not reach the same level of maturity as mathematics, reading, or science, it still provides a meaningful starting point for future research and policy discussions. Ultimately, researchers must approach these data critically, acknowledging both their strengths and limitations. By adhering to rigorous methodological principles and clearly communicating the constraints of the data, analysts can maximize the utility of creative thinking assessments while ensuring that their interpretations remain valid and contextually appropriate. As research in this area progresses, future analyses of the creative thinking data should focus on understanding its limitations, assessing its utility in different educational contexts, and identifying ways to responsibly interpret and apply its findings to inform policy and practice.

Conflicts of Interest

Dr. Leslie Rutkowski is Chair of the PISA Technical Advisory Group. This role is advisory in nature and does not influence the publication or editorial decisions related to this manuscript. All analyses were conducted using publicly available data.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Endnotes

[1] If the researcher is not interested in a partition of variance or predictors at multiple levels, variance estimation via replication methods is an appropriate way to deal with the clustered structure of the data.

[2] Because minimum data quality criteria will vary across purposes, we do not offer specific guidance here.

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